"श्री"

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Lecture 1

Topics :- Introduction to Random Variables
- Indicator Random Variable
- Hiring Problem

* Random Variable

→ Let's say S is the sample space
→ E.g.
Tossing of Coin 2 times, the Sample space will be

$$S = \{H, T\} \times \{H, T\}$$
$$= \{HH, HT, TH, TT\}$$

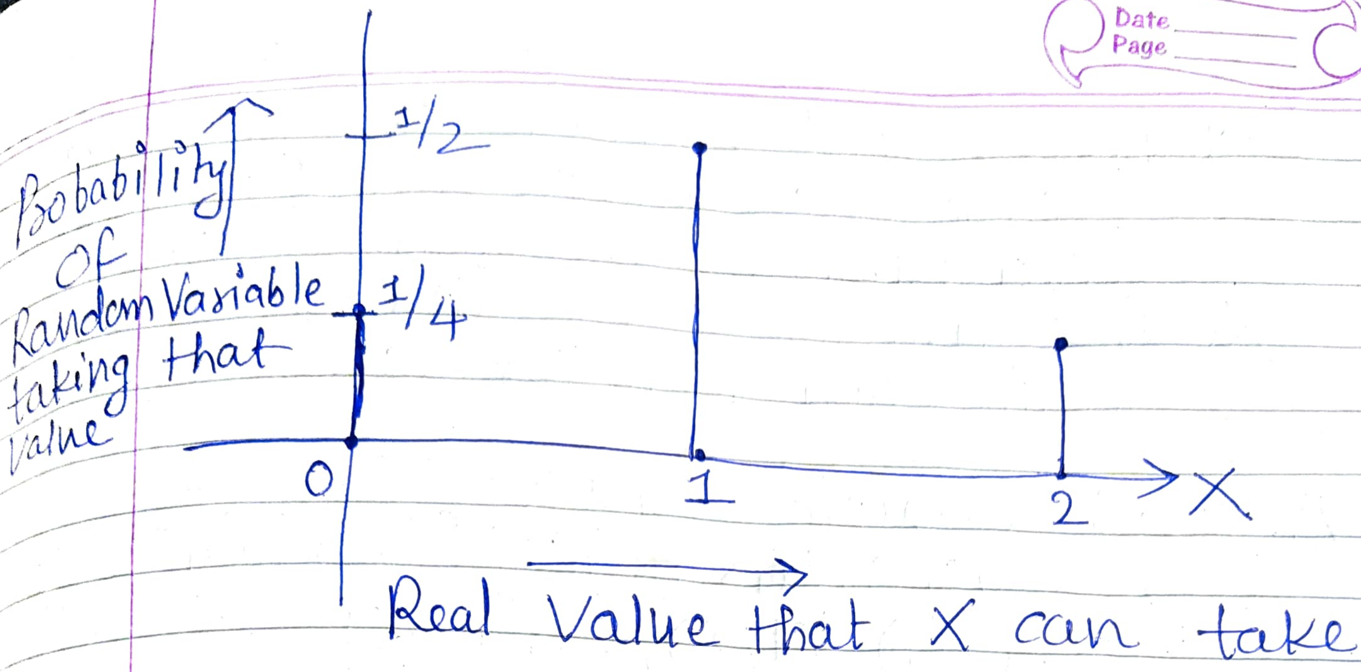
→ Random variable is a functional mapping of each element of S over real space R.

i.e. $X : S \rightarrow R$

X : Random Variable
S : Set of all possible Outcomes i.e. Sample Space
R : Real Value

E.g. $X(\omega)$ = Number of heads in ω

ω	$X(\omega)$: Random Variable
1) HH	2
2) HT	1
3) TH	1
4) TT	0



$$Pr(X=2) = \frac{1}{4}$$

$$Pr(X=1) = \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

$$Pr(X=0) = \frac{1}{4}$$

i.e. Probability that numbers of heads = 2 is 1/4 when we toss Coin twice.

Probability that 1 head is 1/2

Probability that 0 head is 1/4.

→ This is also known as probability distribution of Random Variable.

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Example 2 Let's say we roll a fair die twice and random variable is defined as $X(\omega) =$ Sum of the numbers obtained in each roll for outcome ω.

Answer: Sample Space

$$S = \{1,2,3,4,5,6\} \times \{1,2,3,4,5,6\}$$
$$= \{(1,1), (1,2), \ldots, (1,6),$$
$$(2,1), (2,2), \ldots, (2,6),$$
$$(3,1), (3,2), \ldots, (3,6),$$
$$(4,1), (4,2), \ldots, (4,6),$$
$$(5,1), (5,2), \ldots, (5,6),$$
$$(6,1), (6,2), \ldots, (6,6)\}$$

$$X((1,1)) = 1+1 = 2$$
$$X((1,2)) = 1+2 = 3$$
$$\vdots$$
$$X((6,6)) = 6+6 = 12$$

$\therefore$ X can take values from 2 to 12.

→ We can write the probability that X takes values 2 as

$$Pr(X=2) = \frac{1}{36} \text{ (only 1 case (1,1))}$$
$$Pr(X=3) = \frac{2}{36} \text{ (i.e. (1,2), (2,1))}$$

$Pr(X=4) = \frac{3}{36}$ (i.e. (1,3), (2,2), (3,1))

$Pr(X=5) = \frac{4}{36}$ (i.e. (1,4), (2,3), (3,2), (4,1))

$Pr(X=6) = \frac{5}{36}$ (i.e. (1,5), (2,4), (3,3), (4,2), (5,1))

$Pr(X=7) = \frac{6}{36}$ (i.e. (1,6), (2,5), (3,4), (4,3), (5,2), (6,1))

$Pr(X=8) = \frac{5}{36}$ (i.e. (2,6), (3,5), (4,4), (5,3), (6,2))

$Pr(X=9) = \frac{4}{36}$ (i.e. (3,6), (4,5), (5,4), (6,3))

$Pr(X=10) = \frac{3}{36}$ (i.e. (4,6), (5,5), (6,4))

$Pr(X=11) = \frac{2}{36}$ (i.e. (5,6), (6,5))

$Pr(X=12) = \frac{1}{36}$ (only 1, (6,6))

→ Thus Random variable X can take any value (Real) but probability of X taking specific value is always in the range 0 to 1

✱ Indicator Random Variable (IRV)

Indicator Random Variable for the event A is defined as

$$I(A) = 1, \text{ if Event A occurs}$$
$$= 0, \text{ if Event A doesn't occur}$$

So, it can take only 2 values 0, 1.

E.g. Suppose I toss a coin once
i.e. $S = \{H, T\}$

$$I(\{H\}) = 1, \text{ if Head i.e. } \{H\} \text{ is Obtained}$$
$$= 0, \text{ if Head doesn't appear}$$

→ $I(\{H\})$ can also be written as X_H.

→ X_H : Indicator random Variable X associated with Event Head i.e. H

→ Very important tool to analyze randomized algorithms

☆ Expectation of Random Variable

$$E[X] = \sum_{x} (X=x) \cdot Pr(X=x)$$

Ex 1 E.g. S = Set of all out comes of tossing coin twice

$= \{HH, HT, TH, TT\}$

X : Number of Heads

	$X=x$	$Pr(X=x)$
x	0	1/4
x	1	1/2
x	2	1/4

$$\therefore E[X] = \sum_{x} (X=x) Pr(X=x)$$

$$= 0 * \frac{1}{4} + 1 * \frac{1}{2} + 2 * \frac{1}{4}$$

$$= 1$$

i.e. Expected Number of heads when you toss coin twice is 1.

It is also known as average Value of Random Variable.

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Ex 2 Experiment : Toss a coin thrice
Random Variable X : Number of heads – Number of tails

S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}

Event:	X
HHH	3 − 0 = 3
HHT	2 − 1 = 1
HTH	2 − 1 = 1
HTT	1 − 2 = −1
THH	2 − 1 = 1
THT	1 − 2 = −1
TTH	1 − 2 = −1
TTT	0 − 3 = −3

$$\left.\begin{array}{l} Pr(X=3) = \frac{1}{8} \\ Pr(X=1) = \frac{3}{8} \\ Pr(X=-1) = \frac{3}{8} \\ Pr(X=-3) = \frac{1}{8} \end{array}\right\} \longrightarrow \text{Distribution of Random Variable}$$

$$\therefore E[X] = \sum (X=x)\, Pr(X=x)$$

$$= (X=3) * Pr(X=3) + (X=1) * Pr(X=1) + (X=-1) * Pr(X=-1) + (X=-3) * Pr(X=-3)$$

$$= 3 * \frac{1}{8} + 1 * \frac{3}{8} + (-1) * \frac{3}{8} + (-3) * \frac{1}{8} = \frac{3}{8} + \frac{3}{8} - \frac{3}{8} - \frac{3}{8} = 0$$

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Ex 3 Experiment: Roll a die twice
X: Sum of the numbers obtained in Outcome

$S = \{(1,1), \dots, (1,6),$
$(2,1), \dots, (2,6),$
$\vdots$
$(6,1), \dots, (6,6)\}$

Already found $X = x$ and $Pr(X=x)$ earlier.

$\therefore E[X] = 2 * Pr(X=2) + 3 * Pr(X=3) + 4 * Pr(X=4) + \dots + 12 * Pr(X=12)$

$= 2 * \frac{1}{36} + 3 * \frac{2}{36} + 4 * \frac{3}{36} + 5 * \frac{4}{36} + 6 * \frac{5}{36} + 7 * \frac{6}{36} + 8 * \frac{5}{36} + 9 * \frac{4}{36} + 10 * \frac{3}{36} + 11 * \frac{2}{36} + 12 * \frac{1}{36}$

$= \frac{1}{36}(2+6+12+20+30+42+40+36+30+22+12)$

$= \frac{1}{36} * 242$

$= 6.72$

$$\begin{array}{r} 6.72 \\ 36 \overline{)242} \\ 216 \\ \hline 260 \\ 252 \\ \hline 80 \\ 72 \\ \hline 80 \end{array}$$

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* Few Properties of Expectation :-

1) Expectation is Linear Operator i.e.

$$E[aX+bY] = aE[X]+bE[Y]$$

Here X, Y are Random Variables
a, b are some Scalars.

If $a=1, b=1$ then

$$E[X+Y] = E[X] + E[Y]$$

2) $$E\left[\sum_{i=1}^{n} x_i\right] = \sum_{i=1}^{n} E[x_i]$$

$x_1, x_2, \ldots x_n$ are Random Variables.

$$L.H.S. = E\left[\sum_{i=1}^{n} x_i\right]$$

$$= E[x_1+x_2+\ldots+x_n]$$

$$= E[x_1]+E[x_2]+\ldots+E[x_n] \quad (\because E \text{ is linear})$$

$$= \sum_{i=1}^{n} E[x_i]$$

$$= R.H.S.$$

Thus, $$E\left[\sum_{i=1}^{n} x_i\right] = \sum_{i=1}^{n} E[x_i]$$

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☆ Expectation of Indicator Random Variable

I.V. for Event A $= I(A) = 1$; If Event A occurs
$= 0$; Otherwise

$$E[I(A)] = 1 * Pr(A) + 0 * Pr(A^c)$$

$$= Pr(A)$$

$$\therefore \boxed{E[I(A)] = Pr(A)}$$

Very Important Result for analysis

☆ Suppose you toss a coin once and thus

$$S = \{H, T\}$$

Indicator Random Variable associated with "Head" i.e.

$$I(\{H\}) = 1 \text{ if Head appears}$$
$$= 0 \text{ Otherwise.}$$

Let us denote $I(\{H\})$ as X_H i.e. I.R.V. associated with head when tossing a coin.

$$E[I(\{H\})] = 1 * pr(\{H\}) + 0 * Pr(\{T\})$$

$$= Pr(\{H\})$$

$$\therefore \boxed{E[I(\{H\})] = Pr(\{H\}) = 1/2}$$

Ex: Findout Expected number of Heads when you roll/toss a coin n times.

→ Let X_i denote Indicator Random Variable related to i^{th} toss of a coin.

i.e. $X_i = 1$, if i^{th} toss of coin is Head
$= 0$, Otherwise

→ Therefore we can define Random Variable X,

$$X = X_1 + X_2 + \dots + X_n$$

which denotes total number of heads out of n tosses of a fair coin.

$$\therefore X = \sum_{i=1}^{n} X_i$$

Remember, X_i's are Indicator Random Variables

$$\therefore E[X] = E\left[\sum_{i=1}^{n} X_i\right]$$

$$= \sum_{i=1}^{n} E[X_i] \quad (\because E \text{ is Linear Operator})$$

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→ Recall the Result

E[Indicator R.V.] = Pr of Event Corresponding to Indicator R.V.

$\therefore E[X_i] = $ Pr of Head on i^{th} toss

$= \frac{1}{2}$

→ Substitute this and we will get

$$E[X] = E\left[\sum_{i=1}^{n} X_i\right]$$

$$= \sum_{i=1}^{n} E[X_i]$$

$$= \sum_{i=1}^{n} \frac{1}{2}$$

$$= \frac{1}{2} + \frac{1}{2} + \ldots + \frac{1}{2} \text{ (n times)}$$

$$= \frac{n}{2}$$

$\therefore$ Expected Number of heads $= \frac{n}{2}$
(when we are tossing coin n times)
i.e. on an average we get head half of the times.